



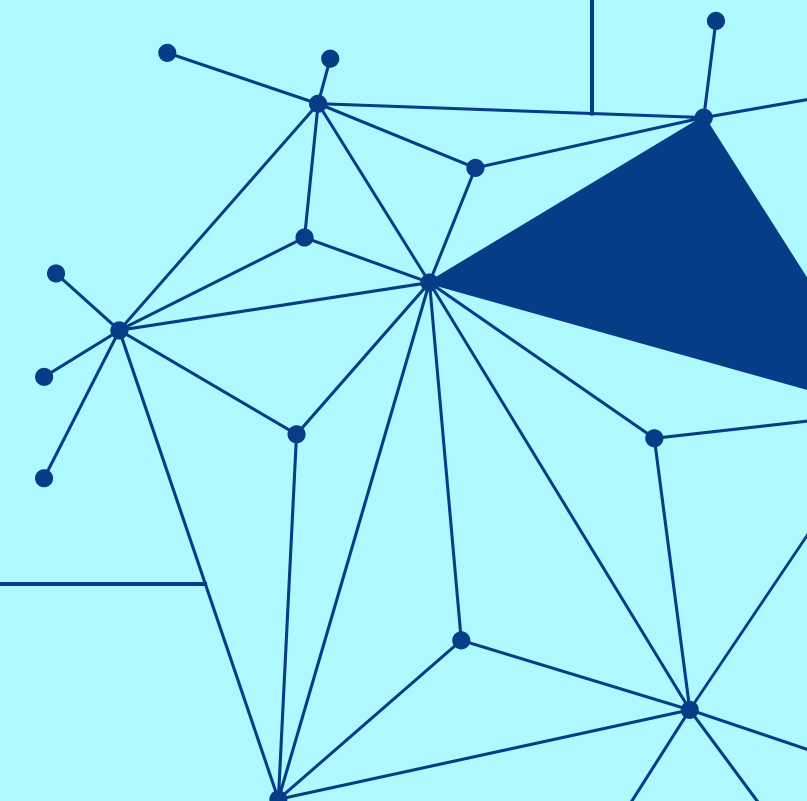
TOPOLOGICAL RESILIENCE AND VULNERABILITY ANALYSIS OF THE DELHI METRO RAIL NETWORK

Group 11

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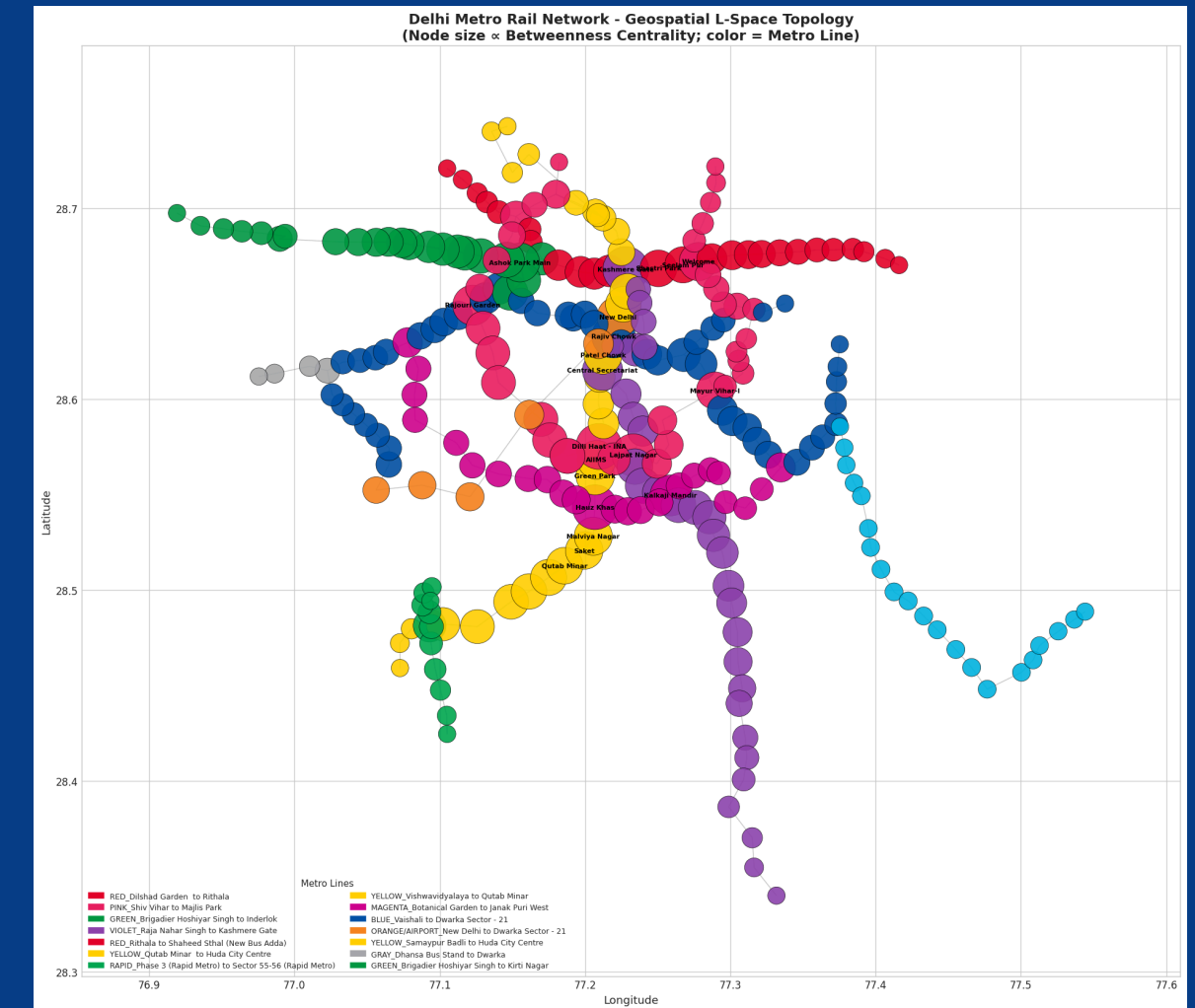


ABSTRACT AND OBJECTIVE

Scope: 262 stations across 36 routes

Goal: Treat DMRC as a weighted spatial graph to uncover hidden structural properties

Key Themes: Dual Topology Construction, Scale-Free Testing, Targeted Attacks and Cascading Failure Simulation, Community Detection, Spectral Graph Analysis



DATA



दिल्ली मेट्रो रेल कॉर्पोरेशन लिमिटेड
Delhi Metro Rail Corporation Limited

GTFS Static Feed: Precise coordinates, routes, and weekday schedules

Wikipedia Scraper: Extracted opening years for 243 stations to model temporal growth



D1: DUAL-TOPOLOGY (L-SPACE VS. P-SPACE)

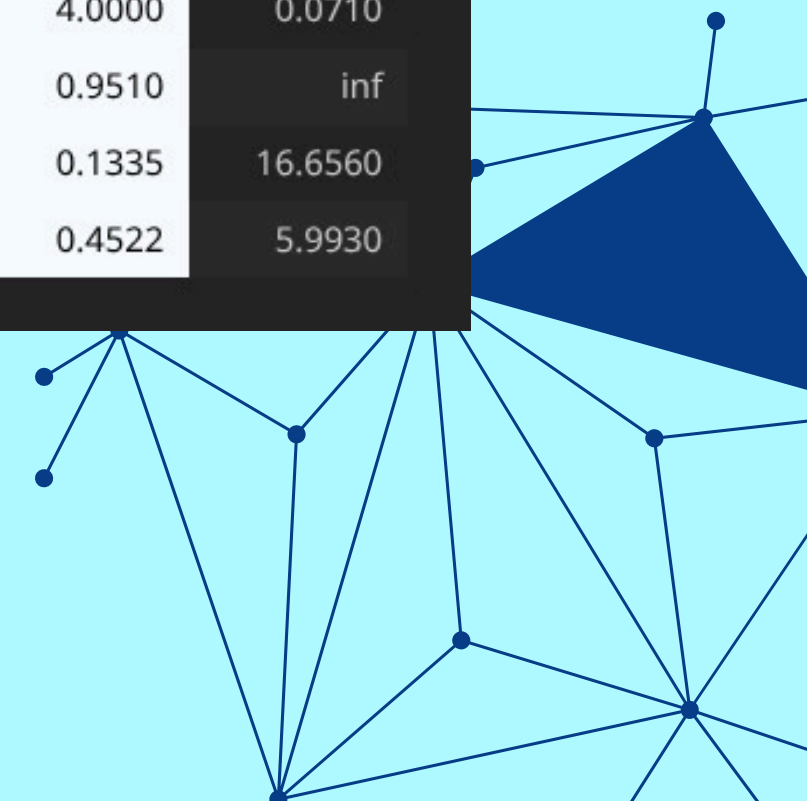
L-Space: Physical adjacency (stops on a track)

P-Space: Functional accessibility (sharing a line)

Result: P-Space path length is **0.12x** of L-Space (8x commuter compression)

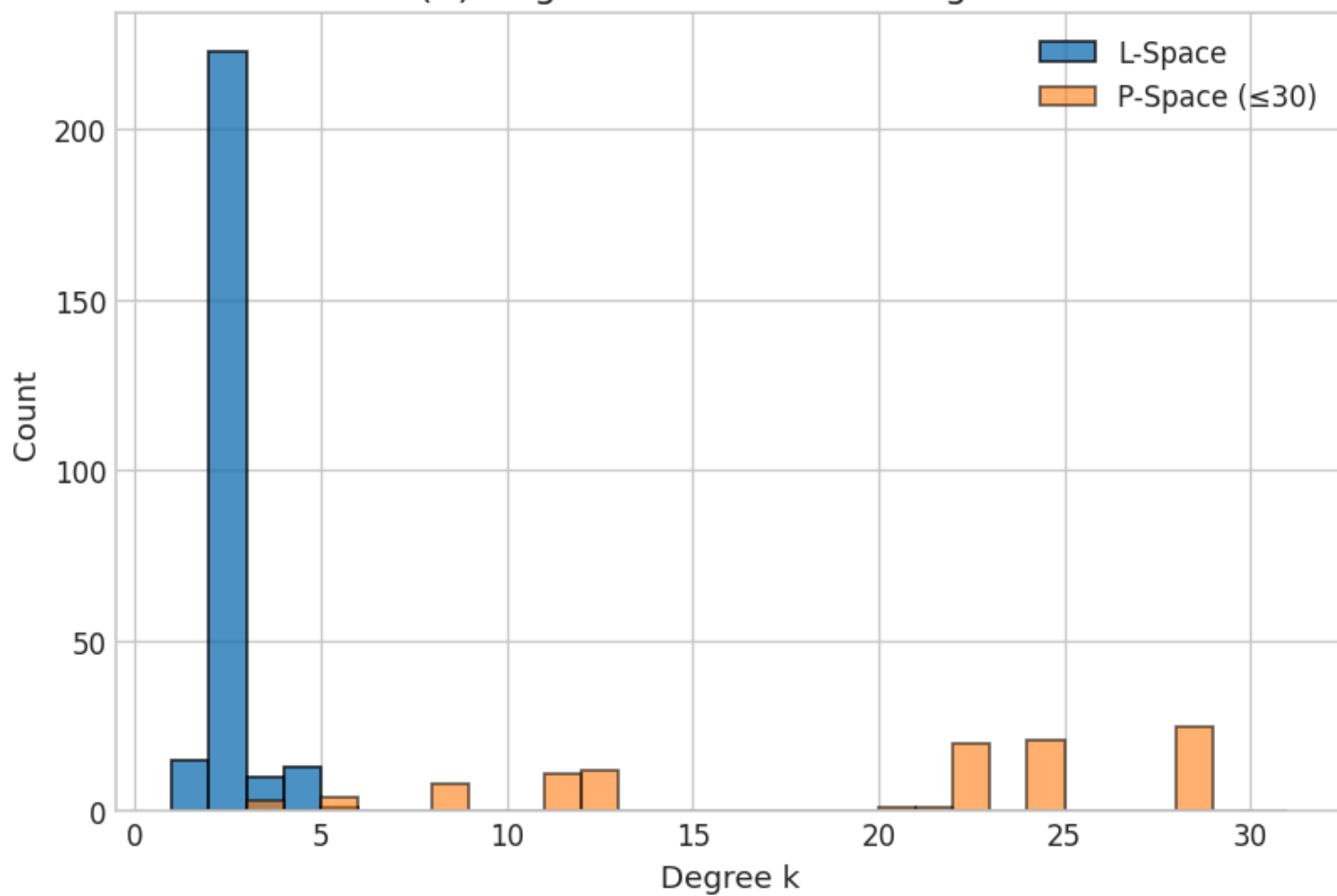
Table 1: Baseline Network Metrics - L-Space vs P-Space

Metric	L-Space	P-Space	Ratio P/L
Nodes (N)	262.0000	262.0000	1.0000
Edges (M)	274.0000	4564.0000	16.6570
GCC Size	241.0000	241.0000	1.0000
Avg Degree $\langle k \rangle$	2.0920	34.8400	16.6540
Max Degree	5.0000	96.0000	19.2000
Avg Path Length L	17.8508	2.1394	0.1200
Diameter	56.0000	4.0000	0.0710
Clustering C	0.0000	0.9510	inf
Density	0.0080	0.1335	16.6560
Global Efficiency Eglob	0.0755	0.4522	5.9930



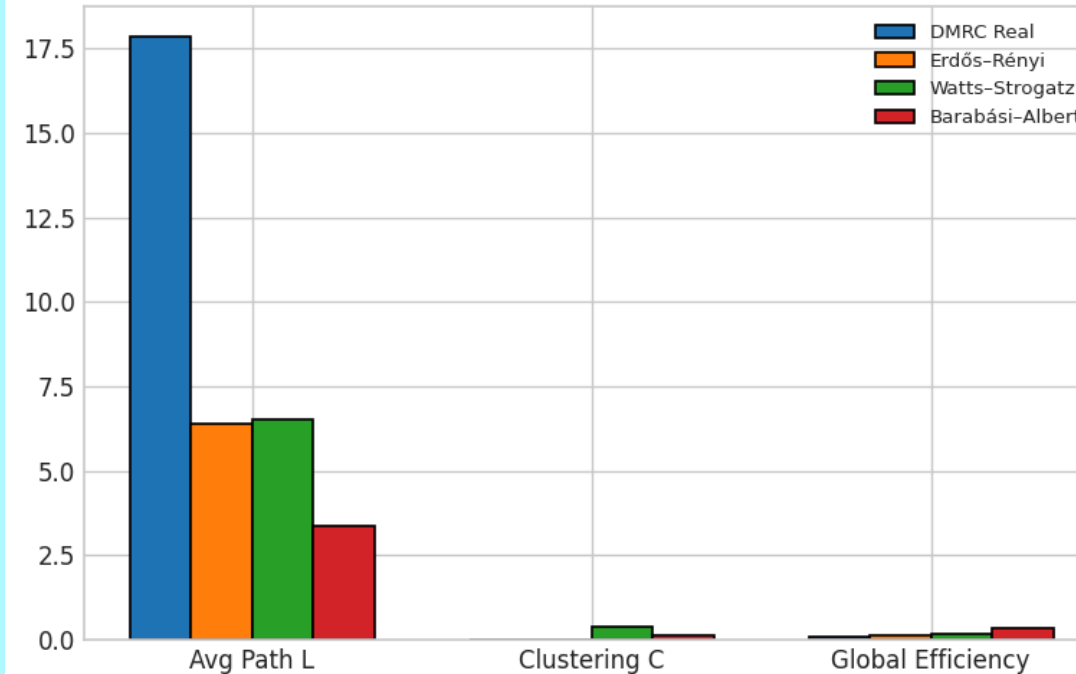
D1: DMRC VS SYNTHETIC NETWORK MODELS

(A) Degree Distribution Histogram

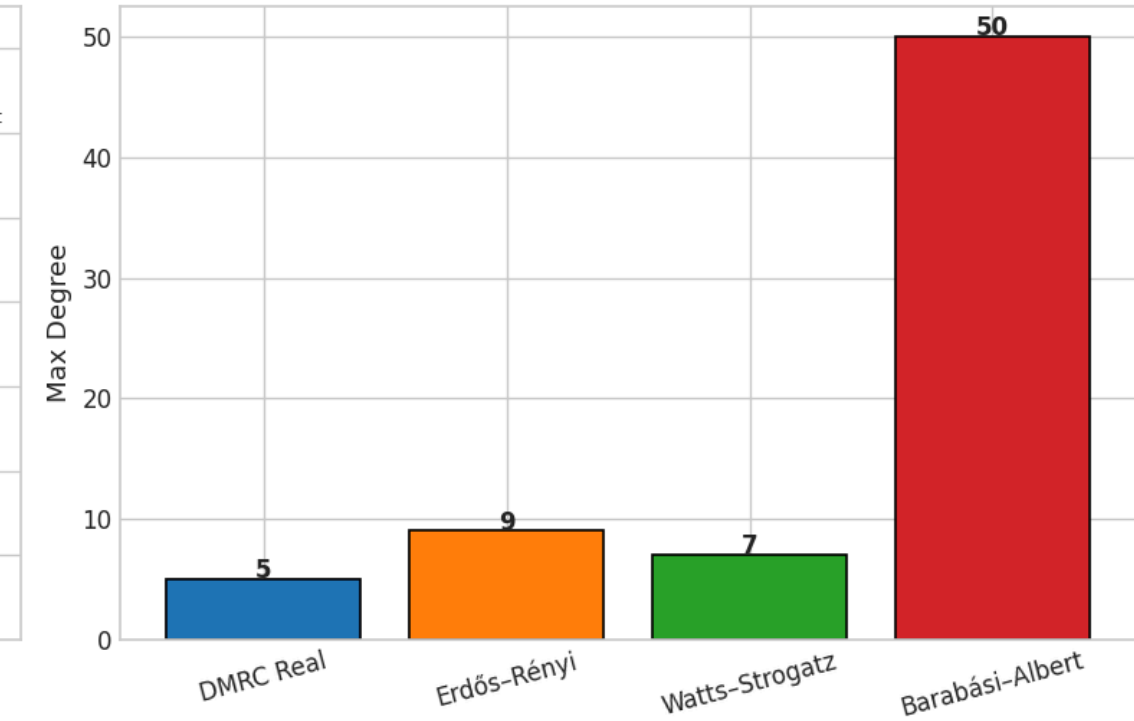


DMRC vs Synthetic Network Models

(A) Structural Metrics Across Models

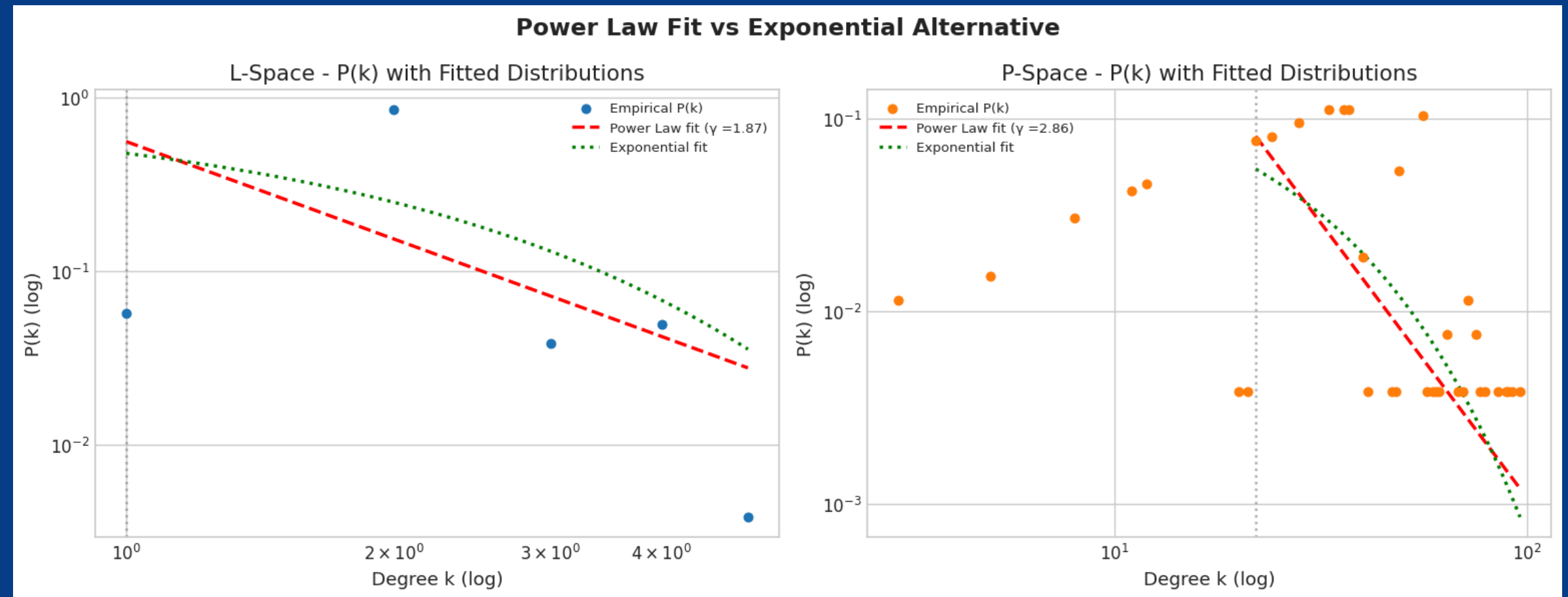


(B) Max Degree - Hub Formation





D2: SCALE-FREE TESTING



Fitted Power-Law Exponent: approx 2.4

Verdict: Not purely scale-free, better described as a "transit-constrained small-world"

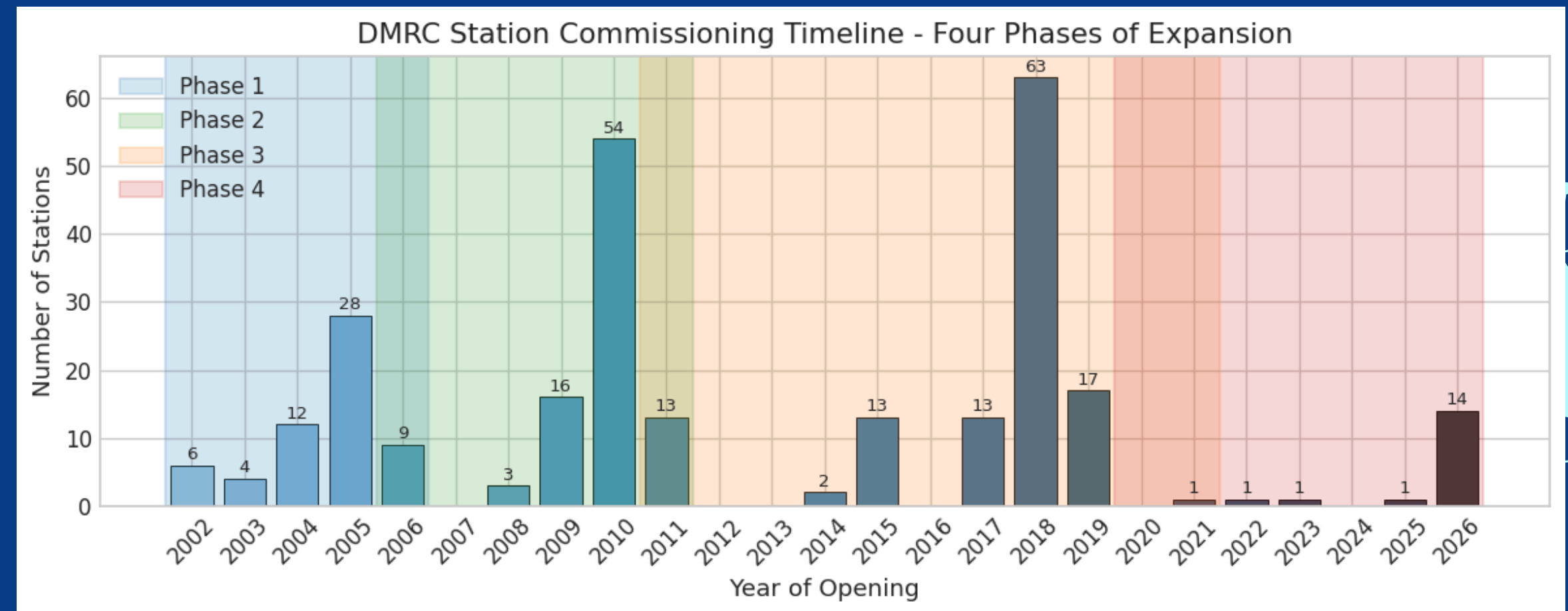
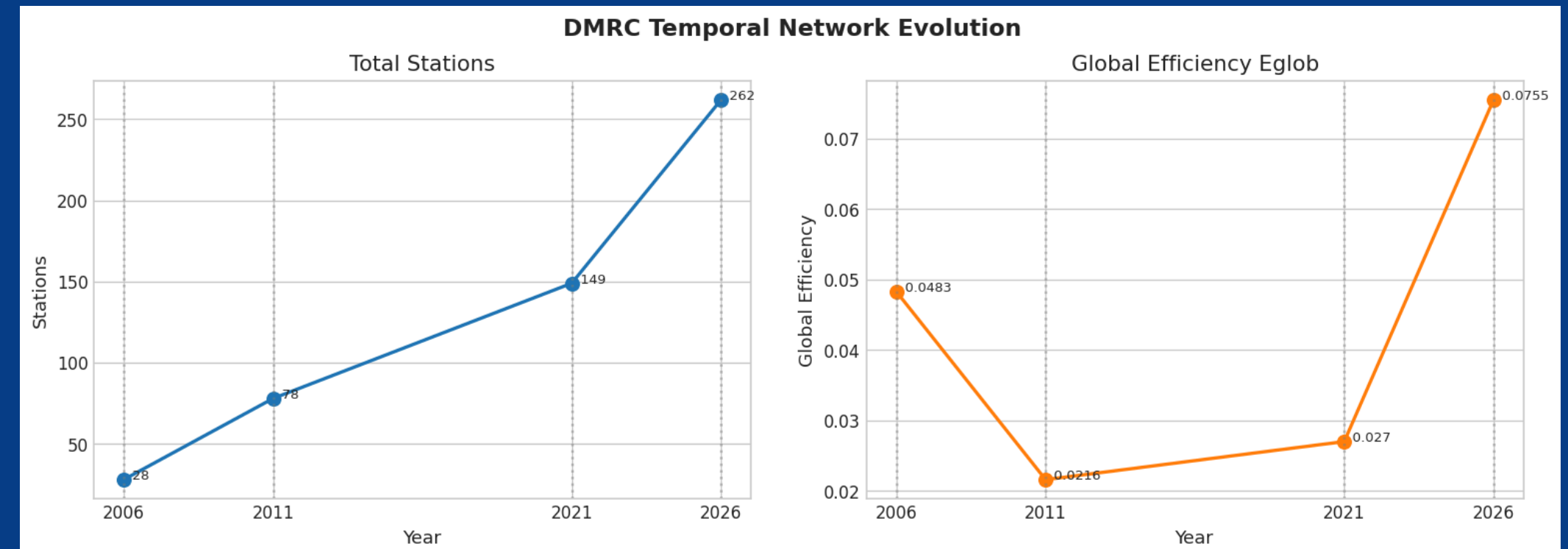
Reason: Hub growth is limited by physical/geospatial constraints



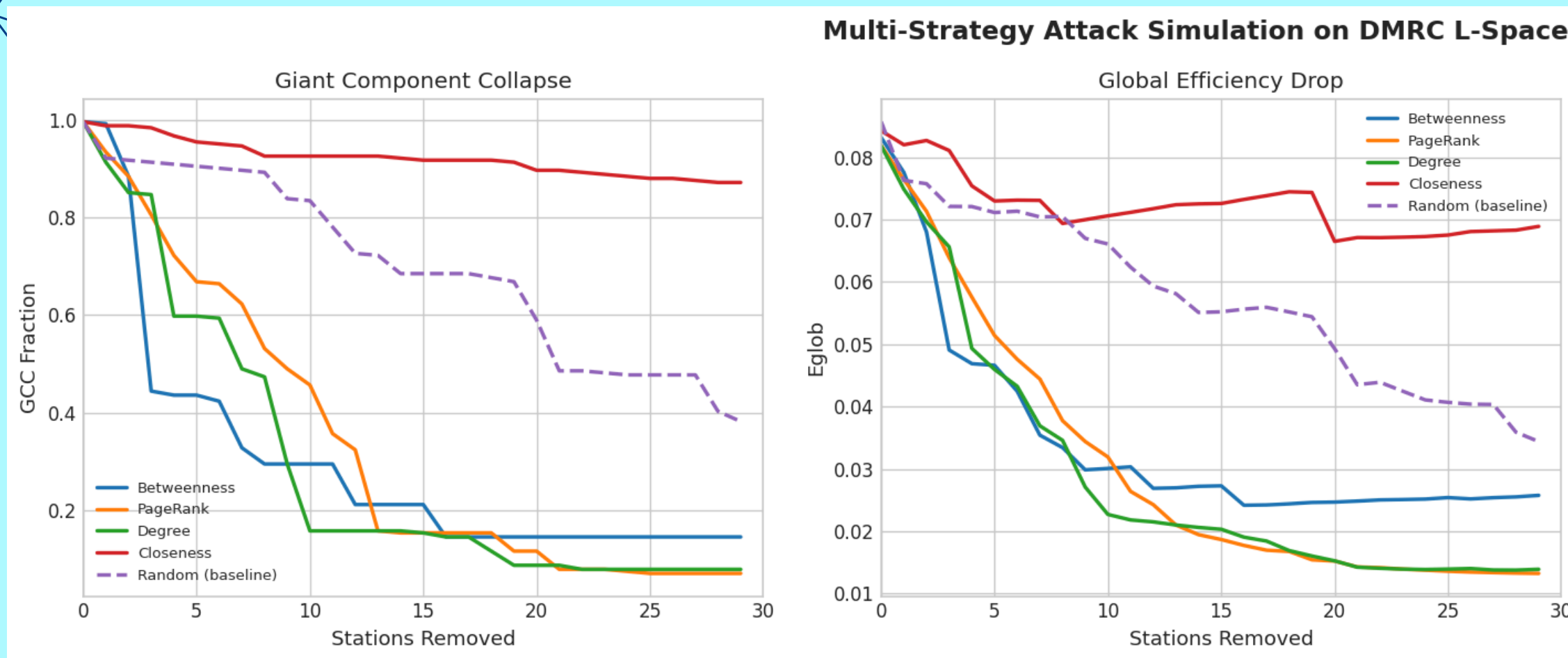
D2: TEMPORAL NETWORK EVOLUTION

Growth Phases: Phase 3 (2011–2021) saw the most aggressive expansion

Trend: Efficiency initially decreased due to rising number of stations but ultimately increased as new interchange hubs reduced path lengths



D3: TARGETED ATTACK SIMULATION



- **Strategies:** Betweenness, PageRank, Degree, Closeness and Random.
- **Key Finding:** Betweenness-based attacks are **3–4x more destructive** than random failures.

Table 8: GCC Fraction Remaining After N Station Removals (by Strategy)

	GCC@5	GCC@10	GCC@15	GCC@20	GCC@25
Strategy					
Betweenness	0.436000	0.295000	0.212000	0.145000	0.145000
PageRank	0.722000	0.490000	0.154000	0.116000	0.075000
Degree	0.598000	0.295000	0.158000	0.087000	0.079000
Closeness	0.967000	0.925000	0.921000	0.913000	0.884000
Random (baseline)	0.909000	0.838000	0.685000	0.668000	0.477000

D3: CENTRALITY RANKINGS

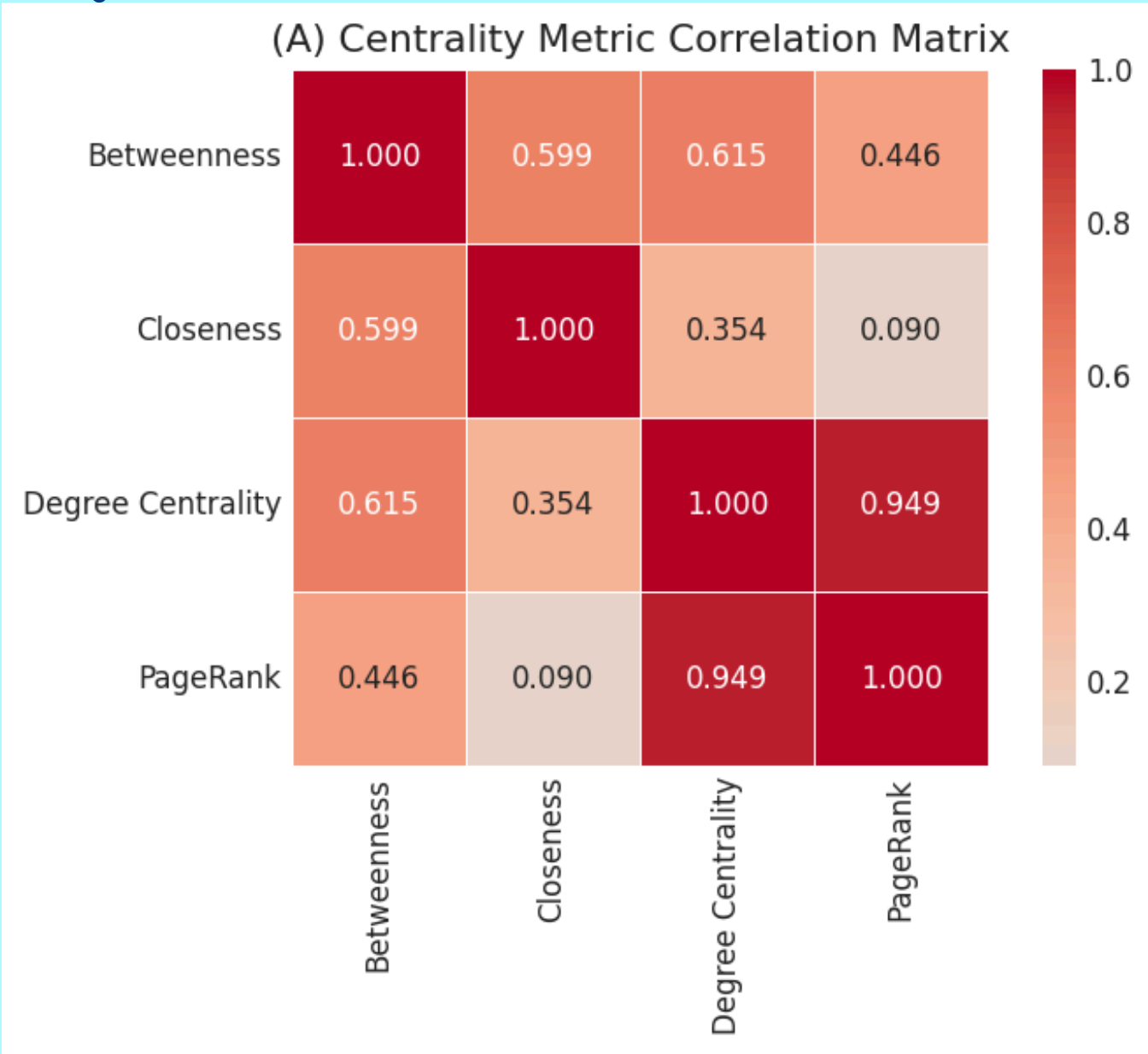


Table 9: Top 15 Stations by Composite Centrality Score

	Betweenness	Closeness	Degree Centrality	PageRank	Composite Score
Station					
Kashmere Gate	0.26187	0.07717	0.02083	0.00832	0.09205
Dilli Haat - INA	0.26328	0.08054	0.01667	0.00676	0.09181
Hauz Khas	0.24822	0.07301	0.01667	0.00696	0.08622
Rajiv Chowk	0.23018	0.08276	0.01667	0.00651	0.08403
Lajpat Nagar	0.21858	0.07856	0.01667	0.00674	0.08014
New Delhi	0.20123	0.08075	0.01250	0.00512	0.07490
Central Secretariat	0.19100	0.08116	0.01667	0.00657	0.07385
Kalkaji Mandir	0.20071	0.06776	0.01667	0.00701	0.07304
Rajouri Garden	0.19160	0.07101	0.01667	0.00695	0.07156
Ashok Park Main	0.17839	0.06691	0.01250	0.00527	0.06577
AIIMS	0.17417	0.07665	0.00833	0.00360	0.06569
Welcome	0.16907	0.06736	0.01667	0.00701	0.06503
Green Park	0.17078	0.07453	0.00833	0.00363	0.06432
Patel Chowk	0.16393	0.08086	0.00833	0.00340	0.06413
Mayur Vihar-I	0.15896	0.07231	0.01667	0.00689	0.06371

D3: CASCADING FAILURE MODEL

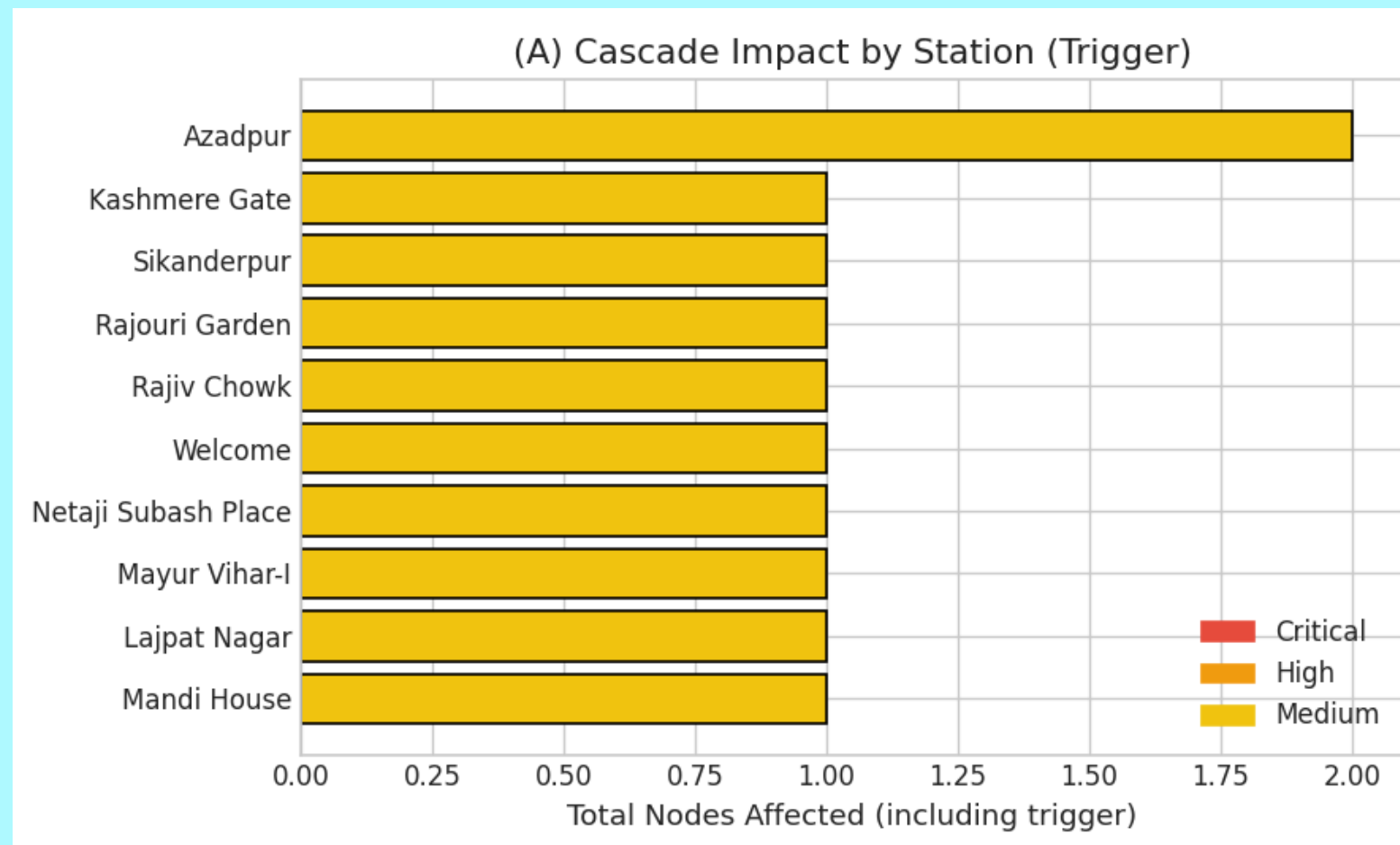
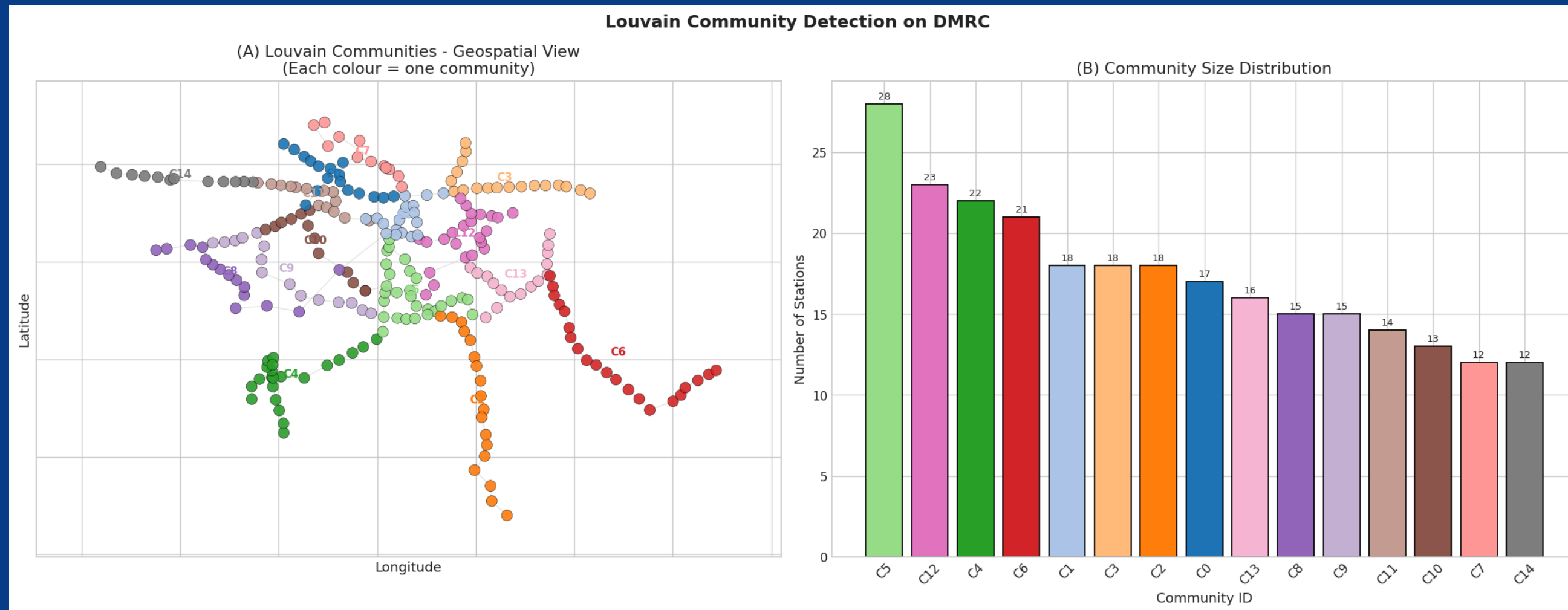


Table 10: Cascade Failure Danger Ranking ($\alpha = 1.5$) - Top 15 Trigger Stations					
Station	Degree	Cascade Impact (Nodes)	Impact %	Cascade Depth	Risk Level
Azadpur	4	2	0.800000	1	MEDIUM
Kashmere Gate	5	1	0.400000	0	MEDIUM
Sikanderpur	4	1	0.400000	0	MEDIUM
Rajouri Garden	4	1	0.400000	0	MEDIUM
Rajiv Chowk	4	1	0.400000	0	MEDIUM
Welcome	4	1	0.400000	0	MEDIUM
Netaji Subash Place	4	1	0.400000	0	MEDIUM
Mayur Vihar-I	4	1	0.400000	0	MEDIUM
Lajpat Nagar	4	1	0.400000	0	MEDIUM
Mandi House	4	1	0.400000	0	MEDIUM
Kalkaji Mandir	4	1	0.400000	0	MEDIUM
Hauz Khas	4	1	0.400000	0	MEDIUM
Dilli Haat - INA	4	1	0.400000	0	MEDIUM
Central Secretariat	4	1	0.400000	0	MEDIUM
Yamuna Bank	3	1	0.400000	0	MEDIUM

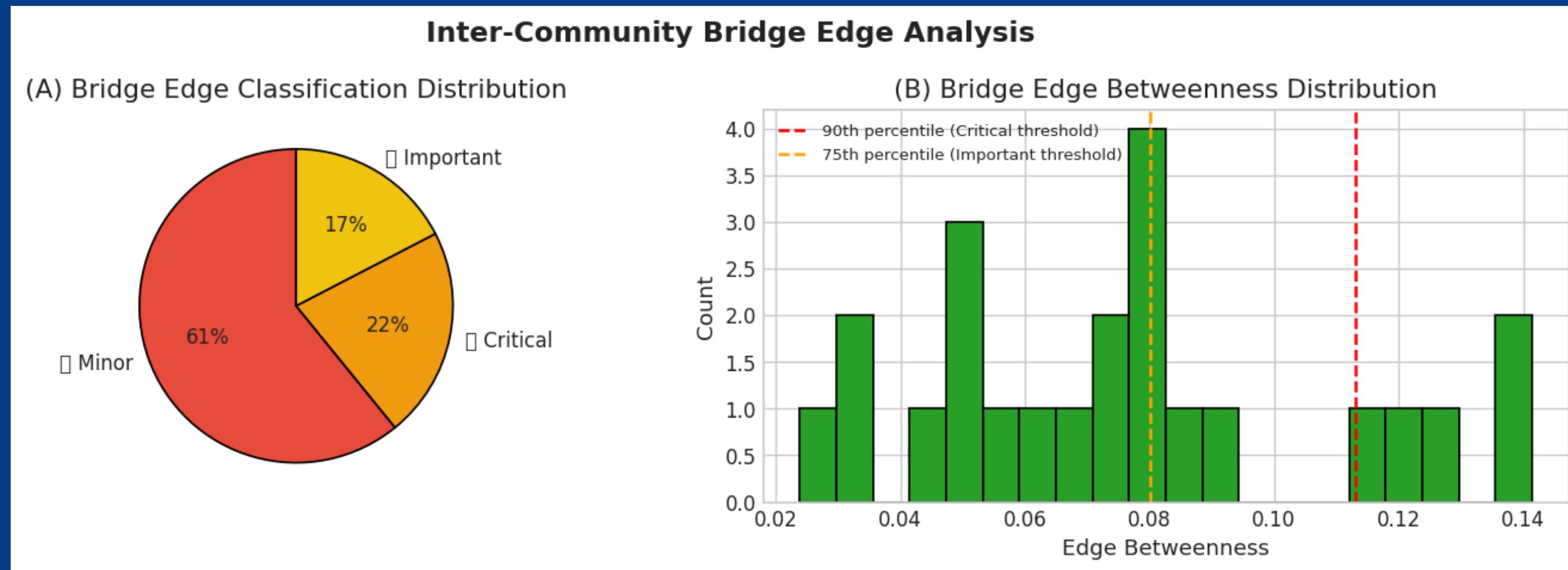
- **Motter-Lai Model:** Redistribution of load with $\alpha=1.5$ (50% overcapacity).
- A single hub closure disables **~15%** of adjacent nodes.

D4: COMMUNITY DETECTION



- Louvain Modularity ($Q = 0.8437$).
- We found **15** geographically coherent communities of sizes
- These emergent communities align with Delhi's **real-world geographic sectors**.

D4: INTER-COMMUNITY BRIDGE EDGES

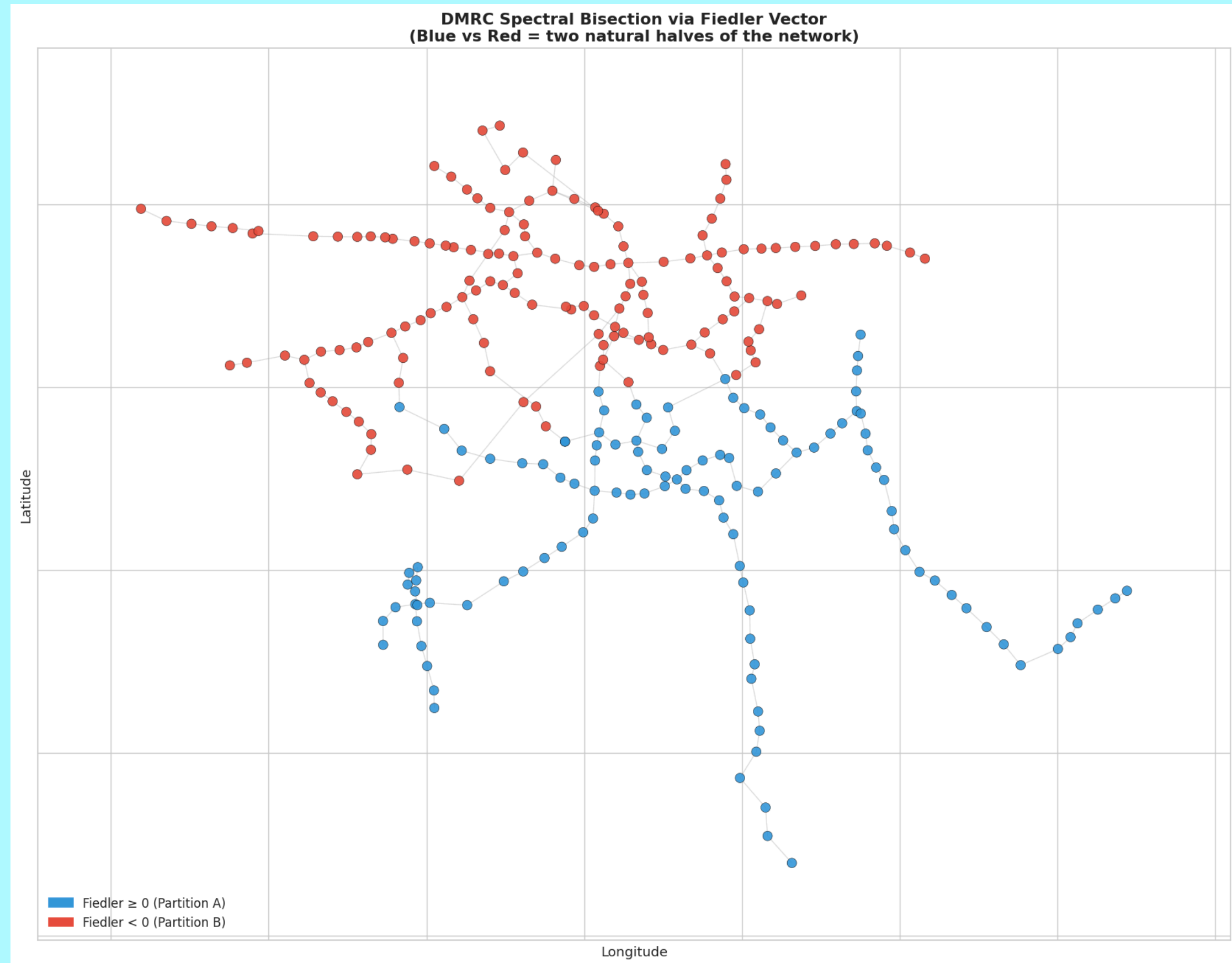


- Found **23 critical** bridge edges
- These edges represent only **~8.4%** of the network but are vital for resilience.

D5: SPECTRAL ANALYSIS

Algebraic Connectivity: P-Space is 172.4x more connected than L-Space

Spectral Bisection: The Fiedler vector naturally divides Delhi into North and South halves



KEY FINDINGS

Theme	Key Finding	Metric
D1: Dual Topology	DMRC is a Small-World network in p-space	$\sigma > 1$ for P Space
D2: Scale-Free	DMRC is NOT purely scale-free; transit-constrained small-world	$\gamma \approx$ fitted
D3: Resilience	Betweenness attack is 3–4× more destructive than random failure	GCC collapse at ~5% removal
D4: Community	Louvain detects geographically coherent communities	$Q > 0.3$
D5: Spectral	P-Space is algebraically more robust; Fiedler vector bisects Delhi N/S	$\lambda_2(P) \gg \lambda_2(L)$

THANK YOU

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